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Studierichting: Informatica
Oefeningenreeks 2E nr. 10.26

$$3 \cdot \left(\frac{27}{8}\right)^{\frac{2x}{3}-1} = 2 \cdot \left(\frac{32}{243}\right)^{2x}$$

$$\Rightarrow 3 \cdot \left(\left(\frac{3}{2}\right)^{3}\right)^{\frac{2x}{3}-1} = 2 \cdot \left(\left(\frac{2}{3}\right)^{5}\right)^{2x}$$

$$\Rightarrow 3 \cdot \left(\frac{3}{2}\right)^{3 \cdot \left(\frac{2x}{3}-1\right)} = 2 \cdot \left(\frac{2}{3}\right)^{5 \cdot 2x}$$

$$\Rightarrow 3 \cdot \left(\frac{3}{2}\right)^{2x-3} = 2 \cdot \left(\frac{3}{2}\right)^{-10x}$$

$$\Rightarrow \left(\frac{3}{2}\right)^{1} \cdot \left(\frac{3}{2}\right)^{2x-3} = \left(\frac{3}{2}\right)^{-10x}$$

$$\Rightarrow \left(\frac{3}{2}\right)^{2x-2} = \left(\frac{3}{2}\right)^{-10x}$$

$$\Rightarrow 2x - 2 = -10x$$

$$\Rightarrow 2x + 10x = 2$$

$$\Rightarrow 12x = 2$$

$$\Rightarrow x = \frac{2}{12} = \frac{1}{6}$$

References